

5.2

Introduction to Data Manipulation in SAP Analytics Cloud

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OBJECTIVE

Use SAP Analytics Cloud (SAC) data exploration and visualization tools to analyze ERP simulation game data.

ACTIVITIES

- Import data.
- Cleanse data and create data models.
- Create data visualizations.
- Share results.

SOFTWARE PREREQUISITES

SAP Analytics Cloud access

DATA REQUIRED

ERPSIM dataset is available in [ERPSIM.xlsx](#)

Scenario

Many courses at SAP University Alliances member schools use the ERP Simulation Game (ERPSIM) to introduce students to integrated business processes using a business simulation game. This game simulates a commodity market wherein teams have to plan, procure, produce and sell products in a competitive environment. The goal is to strategize and run their company to gain the highest

company valuation compared to other teams. The game is played in the SAP ERP system while an external simulator simulates the dynamic market and all the variables that influence the game. By linking SAP Analytics Cloud to the ERPSim instance, this exercise can be used during or at the end of the game to analyze real-time game data. To learn more about ERPSIM, go to erpsim.hec.ca.

Data Visualization

The human visual system has evolved to be particularly good at recognizing patterns. Data visualization has become a standard analytical tool which capitalizes on the ability of humans to recognize patterns within massive quantities of multi-dimensional data generated by business information systems. Many scientific studies have led to the creation of visualization models that utilize human perception and cognition.

When the number of dimensions is small, we can use standard graphing techniques such as bar charts, line charts, histograms, pie charts and scatter plots for visualizations. In this exercise, you will use basic visualization techniques to analyze ERPSim data.

What is ERPSIM?

The ERPSim game is played by teams over several rounds (up to 12 rounds of 20 virtual days each). The teams sell up to 12 products that the market consumes. The products are all muesli cereal in various flavors and box sizes. The teams must forecast demand, run MRP, procure, produce, price, and market their products for sale.

Figure 1 shows the entire cash-to-cash cycle in the game. The transactions in bold are decision points that teams must make and execute. They are considered strategic in nature. The transactions that are gray are considered operational in nature. They are automated by the simulator.

When connected directly to the SAP HANA ERPSim system, teams can run/refresh analytical reports at various points in the game to monitor activities and strategize next steps. For our purposes, we will instead use static data from the results of a simulation game run previously.

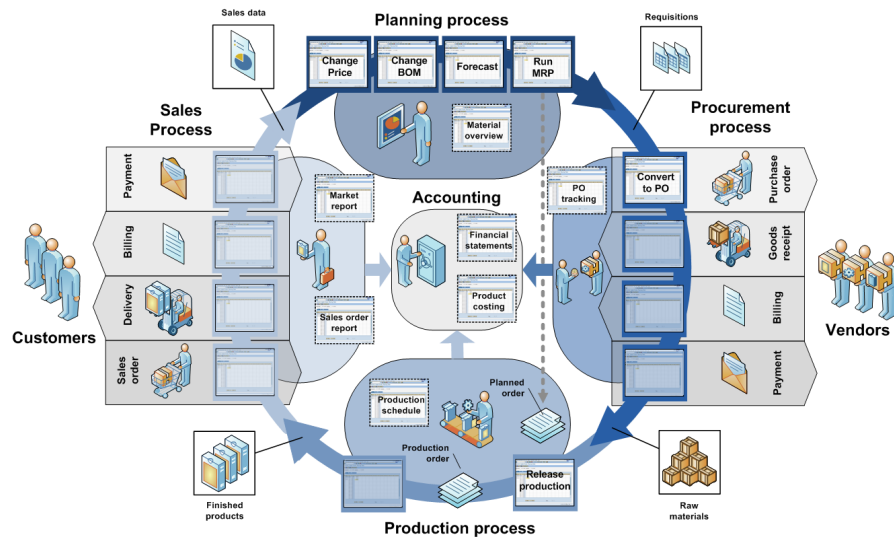


Figure 1: Cash-to-cash cycle

Data Visualization Introductory Exercise

You will now use SAP Analytics Cloud to *acquire* ERPSim game data, and to *visualize* and *discover* interesting trends.

1. Login to the academic tenant of SAP Analytics Cloud (SAC).
2. In the Home screen shown in Figure 2 (yours may look a bit different than mine) expand the menu by clicking on the three horizontal lines to the left of the SAP logo to expand the menu. The functions on your menu are determined by your security authorizations. Take some time to explore the menu items in SAC.

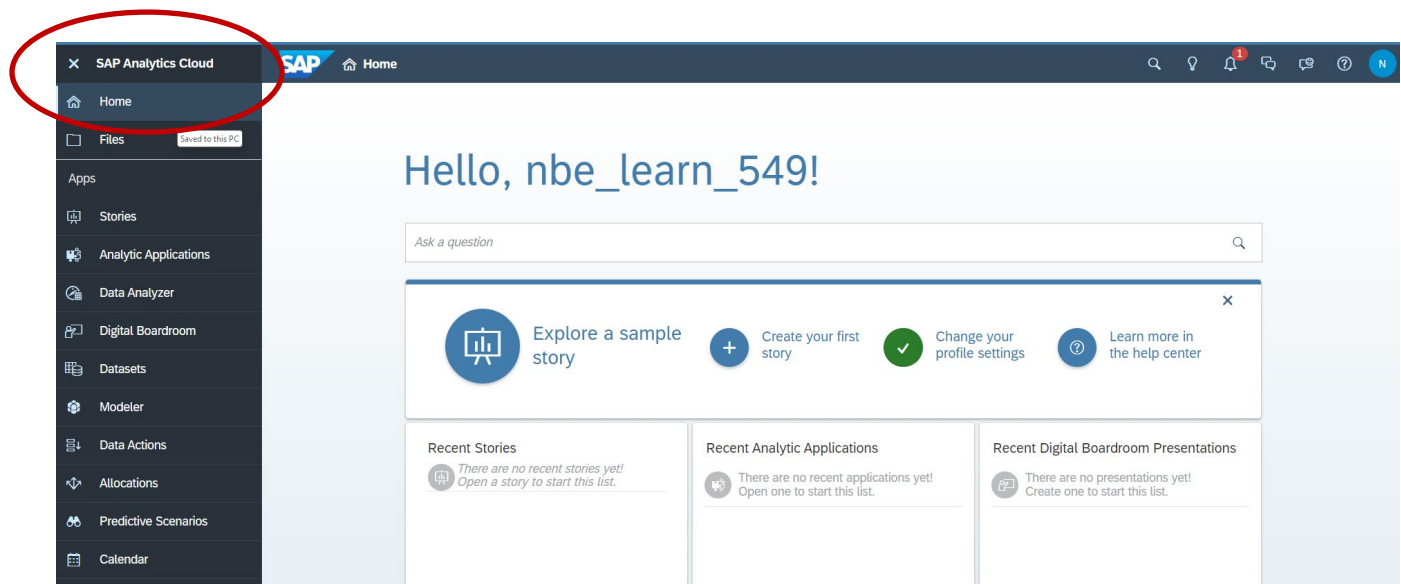


Figure 2: SAP Analytics Cloud Home Page

3. Visualizations are created in SAC using Stories. We will start by creating a story and importing data. There are multiple methods to create stories. The following is one way to do it.
 - a. Click Stories. You will now see multiple options for creating a story. Responsive design allows the charts to resize depending on the screen they are being displayed on. Canvas is for “pixel-perfect” visualizations. Grids are self-explanatory and From a Smart Discovery is the result of SAC’s built-in intelligent analysis capabilities.
 - i. Choose Create New Canvas.
 - ii. When the system asks you to select the Design Mode Type, choose Classic Design Experience.
 - iii. Create.
 - b. You will now see a page asking you to Add data and Add an object to the canvas. Add data.

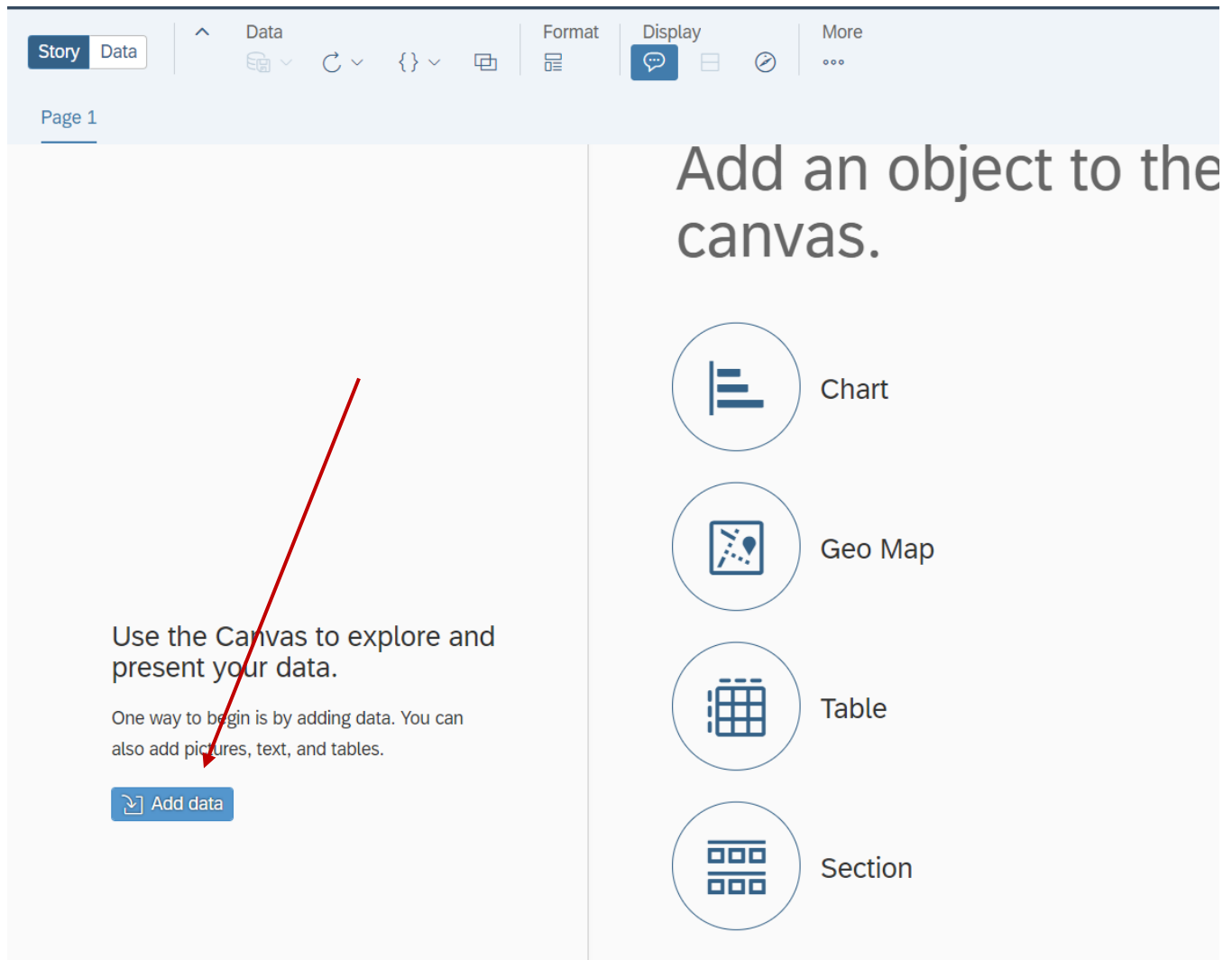


Figure 3: What the Screen looks like when you Create a New Story

- c. Choose the Data Uploaded from a File icon.

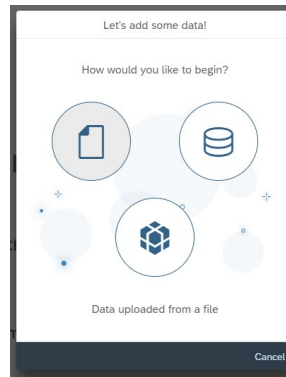


Figure 4: Selecting a File to Explore

- d. Select Source File and find and choose the *ERPSIM.xlsx* file, (you should have downloaded this file to your device).
- e. Open.
- i. “Use first row as column headers” should be checked.
- f. Import. Wait for the data to load into SAC. You may get a message stating that “because there are large number of rows in your data set, we have selected a sample for you to work with. Any work done on the sample will be applied to the full data set at model creation.” If so, click Got It.
- g. You will see a Data view of your data. Notice that the system identified variables in the data file as measures and dimensions.

What is a *measure*?

A measure is a field on which calculations can be made. These are fields of business interest for analytics such as revenue, profit, quantity sold. The calculations can be sum, min, max, average, count, etc. Measures are also called *key figures* or *facts*.

What is a *dimension*?

A dimension is reference information about a measure. It provides context for the measures; for example customer, time, or product. *Revenue by customer* is an example of how you would report a measure by a dimension.

- h. Click on the Team attribute. You see it is identified as *Dimension*, correctly.

- i. Click on the Round attribute or identify the icon in the column header. Because it is a numeric value, SAC has identified it as a measure.
- i. Change Round to a *Dimension* by clicking on the More menu of Round and choosing Change to Dimension from the dropdown box as shown in Figure 6.

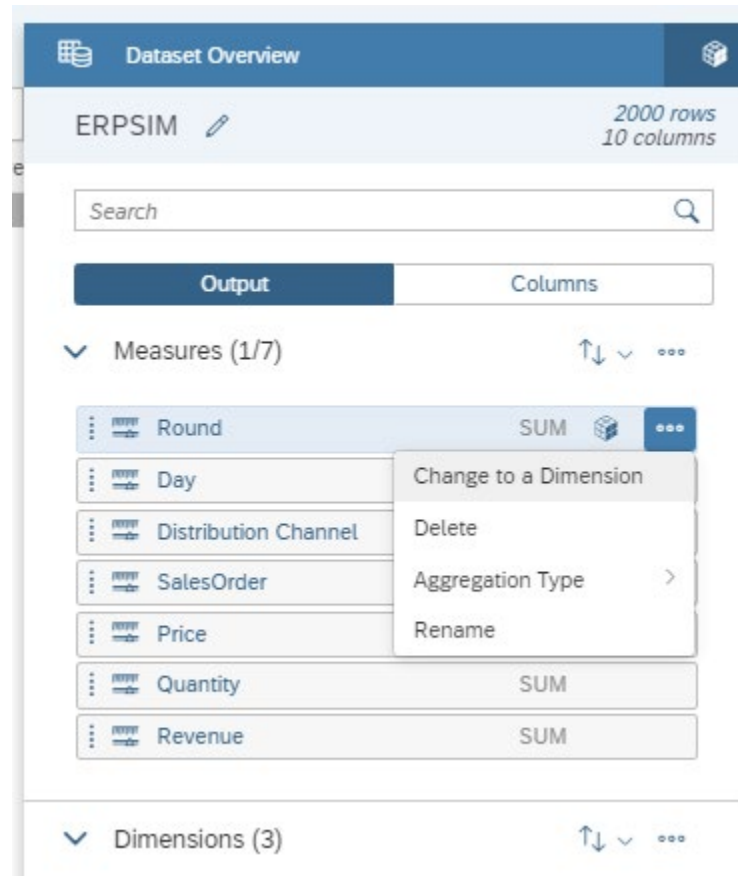


Figure 5: Dimensions and Measures

- j. Similarly, *Day*, *Area*, *Distribution Channel*, and *SalesOrder* should all be Dimensions. If they were incorrectly identified as measures, change them as described in the previous step.
- k. *Price*, *Quantity*, and *Revenue* should be the only Measures. There will be 3 Measures and 7 Dimensions in the ERPSim data.
4. Enhancing the data.
 - a. The values in some of the Dimensions are not particularly descriptive so you will want to follow these instructions to make them more meaningful.
 - b. Highlight the *Area* column.
 - c. Click on the downward-pointing arrow to show additional actions.

- d. Select Create a Transformation → Replace. (See Figure 7 for the location of the Transformations icon.)

AA Area	22 Distributio...	22 SalesOrder	AA Product	1 st Price	22 Quantity	1 st Reve
WE			500g Mixed Fruit	5	3242	16210
NO			500g Original Mue	5	4000	20000
SO	14	0000004342	500g Blueberry M	5.5	3196	17578
NO	14	0000004338	500g Blueberry M	5.5	3149	17319
SO	14	0000004334	500g Mixed Fruit	5	3124	15620
WE	12	0000007313	500g Strawberry I	4.2	7587	31865
NO	14	0000004329	500g Nut Muesli	5	3626	18130
WE	12	0000004323	500g Blueberry M	5.25	5463	28680
WE	14	0000004277	500g Mixed Fruit	5	3766	18830
WE	14	0000004311	500g Mixed Fruit	5	3421	17105
SO	14	0000004346	500g Nut Muesli	5	3795	18975
WE	12	0000004300	500g Original Mue	4.25	8163	34692
NO	12	0000004292	500g Original Mue	4.25	7829	33273
NO	12	0000007329	1kg Strawberry M	4.75	7776	36936

Figure 6: Transforming Data

- e. Replace NO with **North**, SO with **South** and WE with **West**. This can be done simply in SAC by entering the values between the quotation marks. An example is shown below in Figure 8. Hit enter on your keyboard or click on the check mark to save the transformation.
- i. Be sure to replace all three values, NO, SO, and WE.

AA Team	22 Round	22 Day	AA Area	22 Distributio...	22 SalesOrder
1	KK	4	WE		0000004306
2	KK	4	North		0000004343
3	KK	4	SO	14	0000004342

Figure 7: Transforming Data using the Replace Function

- ii. You will now see the transformed data in the Area column.
- f. Select the *Distribution Channel* column.
- i. Choose the Details panel.
- ii. Change the Data Type to String, Statistical Type to Textual, (Figure 9).

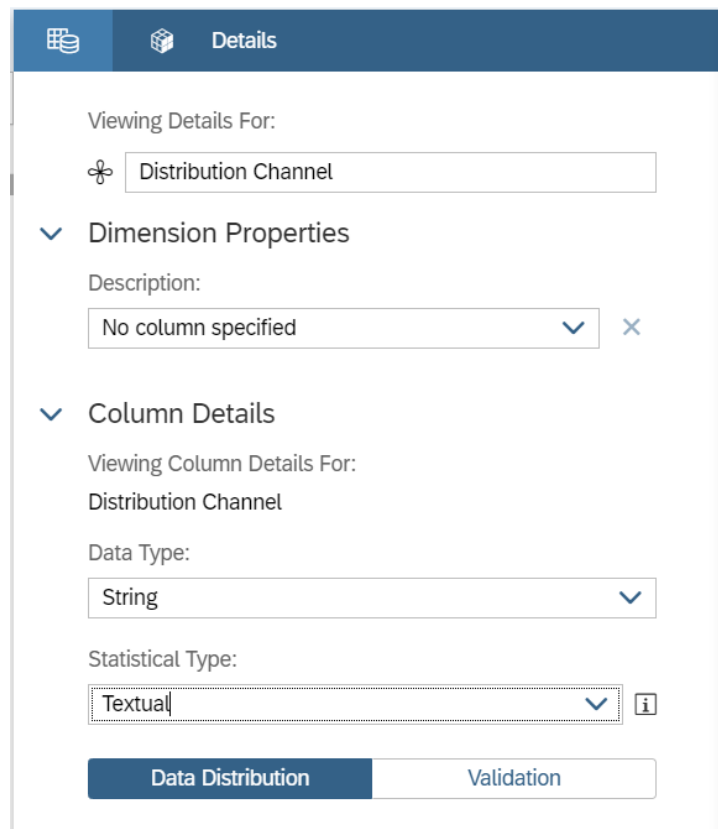


Figure 8: Change Dimension Data Type

- g. Transform the Distribution Channels – 10: [Hypermarkets](#), 12: [Grocery Chains](#), 14: [Convenience Stores](#) using the Replace transformation.

Question 1. Examine the Distribution Channel Details panel (to the right of the screen). Which Distribution Channel has the highest number of sales?

- h. It might also be interesting to look at the data not only by *Round* or *Day*, but by a combination of *Round* and *Day*. One way to do this is to create a hierarchy.
- Use the shift key to highlight both *Round* and *Day* columns.
 - Select the Level-based Hierarchy icon from the Actions section of the ribbon.

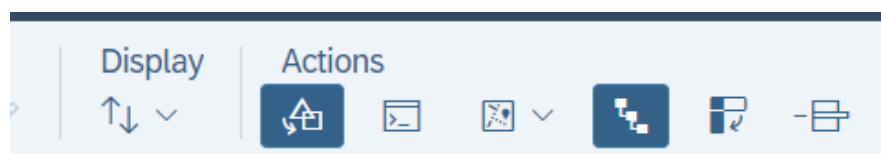


Figure 9: Actions Ribbon

- Name your hierarchy.
- Create the levels of *Round* and *Day* by selecting the Dimensions as shown in Figure 10.

Hierarchy

Name: RoundDay Hierarchy

Levels:

- Round
4, 7, 3, 8, 5, 6, 2, 1
- Day
12, 13, 23, 11, 29, 14, 21, 16, 15, 22, 7, 8, 25, ...

Select Dimension

OK Cancel

Figure 10: Round Day Hierarchy

- v. OK.
- i. Highlight the *Revenue* column and scroll down on the Details panel. You should see the histogram visualization in Figure 11. Hover over the columns to view the statistical measures of each bin.

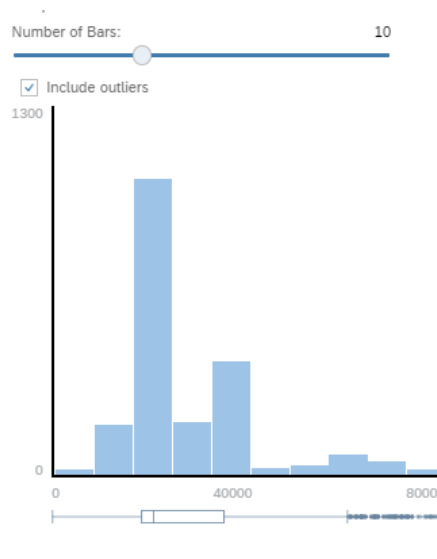


Figure 11: Exploring the Revenue Data

- j. Explore the other measures of the data set.
5. Validate Full Dataset to check for data issues. There should be none.
 6. Save. You will receive a warning that the system will need to save the modifications to the model. Save. During the conversion, you may also see some messages about data types that the system detected and changed to the correct type. This is a nice feature of the system that it checks our work and fixes what we missed.
 - i. Choose your folder and name the file “*ERPSIM_YourName*”.
 - ii. Enter your name as the description.
 - b. *You have just created what is called an “embedded” or “private” model in SAC. Embedded models are specific to the story and are not shared publicly. They are a type of dataset.*
 - c. Click OK.
7. Go to the Story view and click on Run a Smart Discovery to add an automated discovery. You can see from Figure 12 that there is more than one way to add a Smart Discovery.

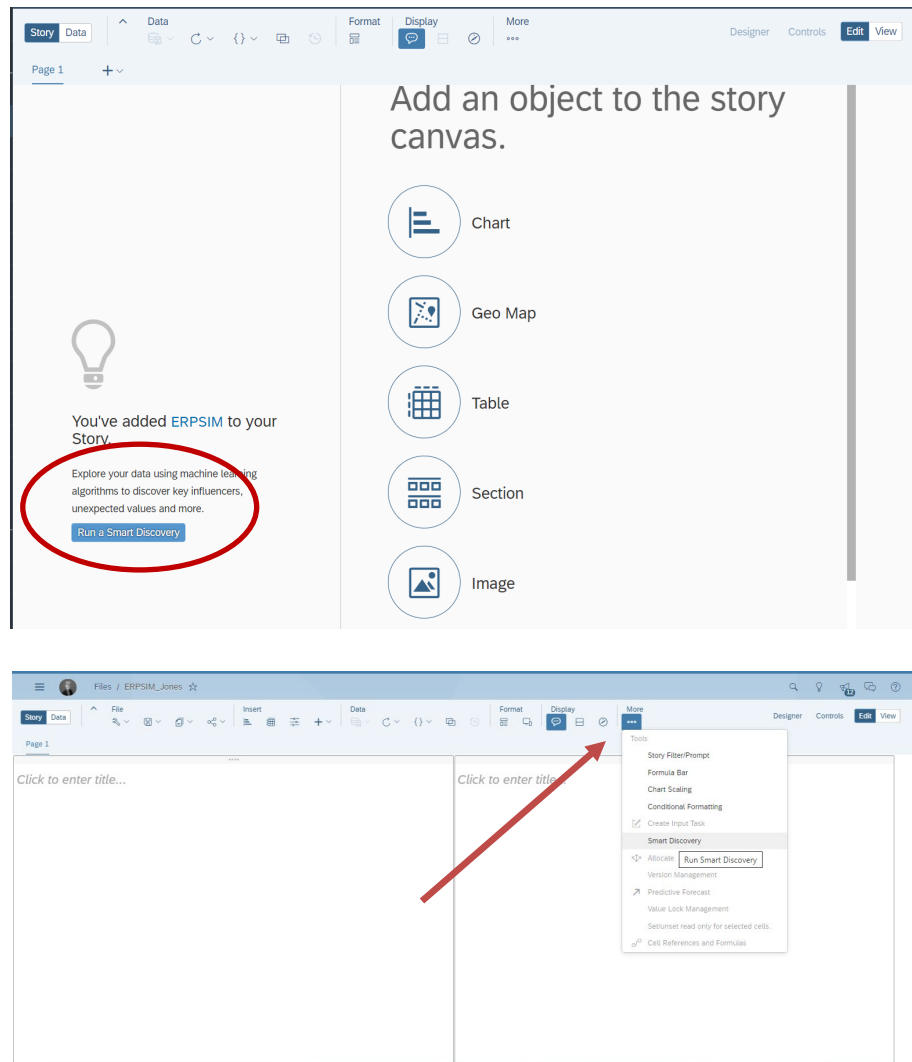


Figure 12: New Smart Discovery

- On the Smart Discovery panel, select Revenue as the Target and select all of the dimensions EXCEPT Day and Round as Entity.
- Run.
- SAC will create four pages of visualizations. Answer the following questions using the smart discovery results.

- Question 2. Which combination of Entities had the highest revenue from sales of muesli?
- Question 3. What was the highest sale? (*Hint: look for max.*)
- Question 4. What are the key influencers of Revenue? Is this what you would expect? Why or why not?
- Question 5. Notice that the key influencers are separated into assigned bins. Bins are ranges of values and the frequency of occurrences within those ranges is

- recorded as a histogram. Identify the most commonly occurring Revenue and Price bin. What are the values in the bin? What is this telling you?
- Question 6. Which price bin impacts average Revenue the most? Why does this make sense?
- Question 7. Which Distribution Channel has the highest Average Revenue? Why use average revenue instead of total revenue?
- Question 8. Examine the simulation page. How does changing the team affect the Expected Revenue? What are the effects of changing Price? Area? Distribution Channel?

8. Creating additional visualizations for the story.

- a. Add a Page by clicking on the + sign to the right of the Simulation page.

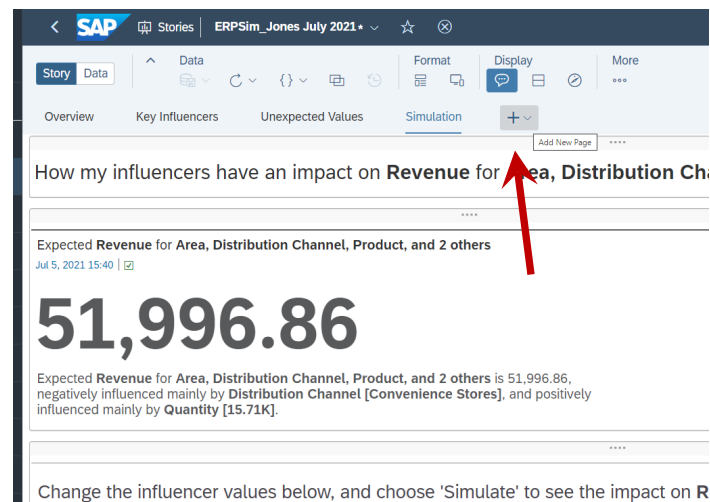


Figure 13: Adding a Page to the Story

- b. Select Canvas.
- c. Select Chart as the object to add to the story.
- d. The Chart canvas consists of the canvas workspace on the left/center of the screen and the Builder and the Styling panels on the right. Use the Builder to create charts that answer the questions highlighted in gold. Be sure to use the appropriate charting technique and good charting practices. Hints are provided for most questions.
- e. You may choose to add more than one chart to each page or you may add a new page to answer each question. To add a chart to the same page, click on Insert → Chart. *(Hint: You can rearrange charts on a page by grabbing them and moving them to wherever you want them to be.)* To add a page, hover your mouse to the right of the last page and you will see a + sign. Click on the + and click on Canvas to get a new canvas page.

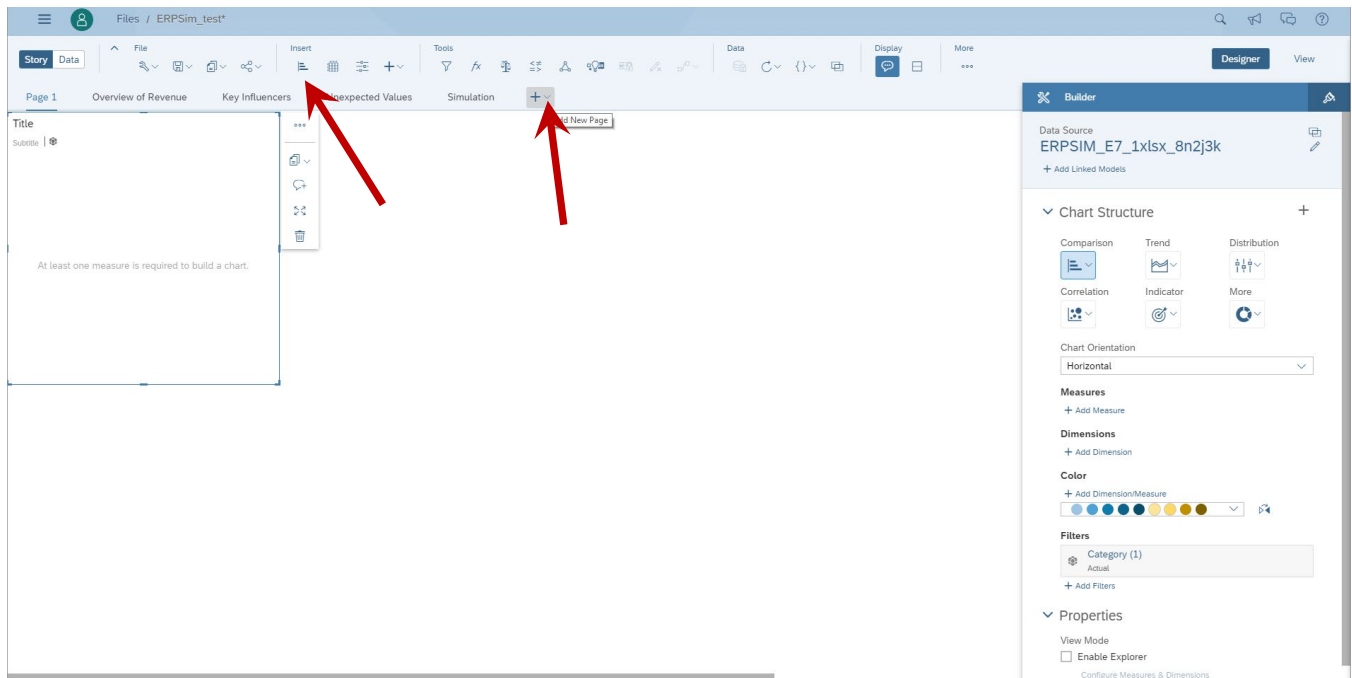


Figure 14: Adding a Visualization to the Story

- i. You may choose to rename each of your additional pages rather than leaving them as 1, 2, 3 and so on by clicking on the dropdown arrow next to the page name, see Figure 15. *Hint: You might have to hover a bit to the right of the page name to get the page menu dropdown to show.*
- ii. You may also reorder the pages by dragging them to where you want them in the order.

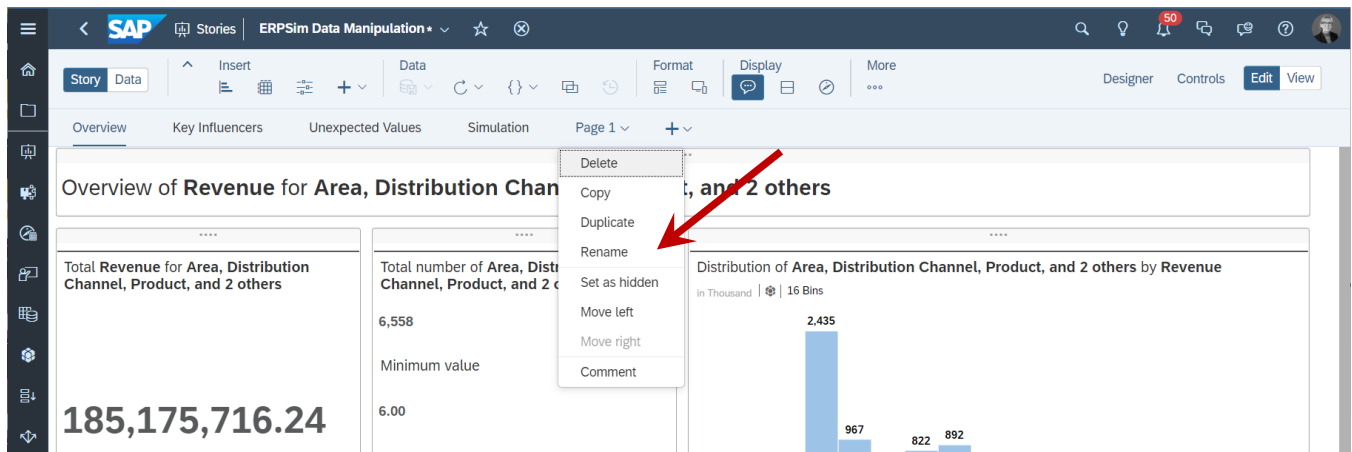


Figure 15: Rename the Story Pages

Note: There is more than one way to accomplish your charting tasks in SAP Analytics Cloud. The developers have built access to the many tools and functions in multiple locations within the

canvas and on the tabs. Explore and find the way that you prefer. The Hints are exactly that, “hints” and should not be construed as the only way to create the appropriate charts.

Question 9. Which team had the highest revenue in total? What was the revenue?

Hint: Use a *column* chart. MEASURES: *Revenue*. DIMENSIONS: *Team*. Choose any color. Change the chart orientation to *Vertical*. Click on the Chart and in the properties, sort Revenue *Highest to Lowest*.

To enable interactivity in the Digital Boardroom, activate the Boardroom Properties as shown in Figure 21. You will also want to enable Explorer at the bottom of the Builder menu. Do this for all of your charts.

Include your name in the title of every chart. (Double click on the chart title to rename it.)

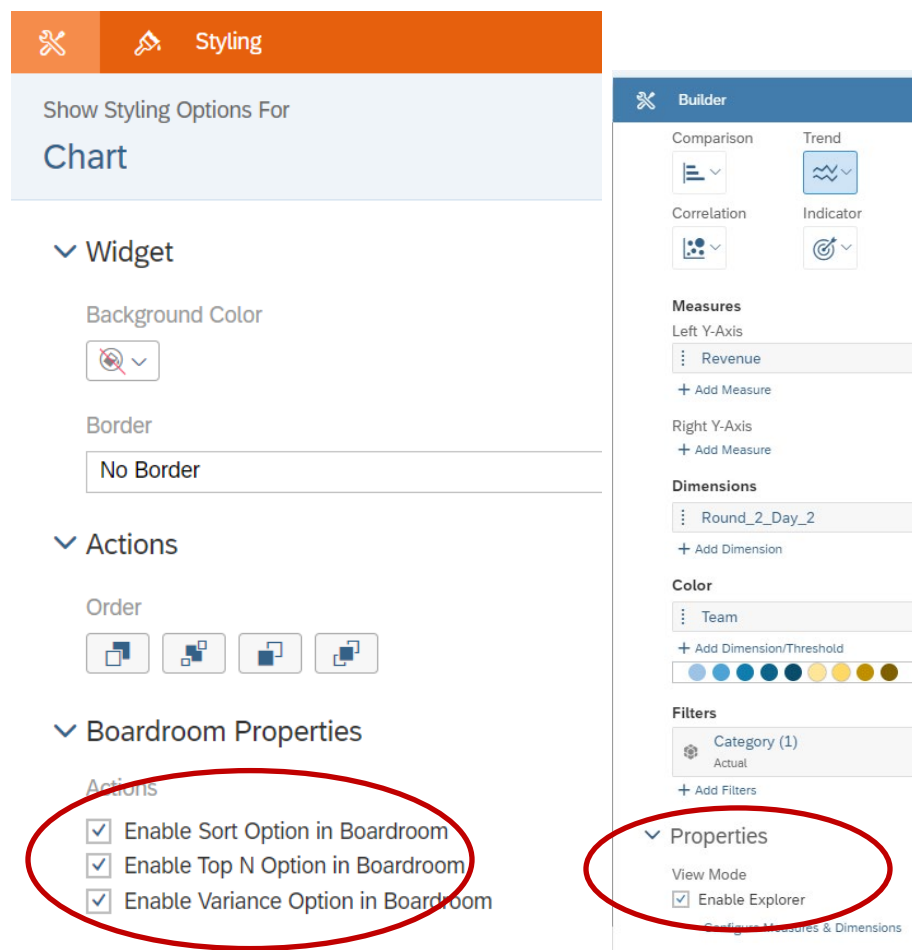


Figure 16: Working with Charts

Question 10. What product had the highest revenue in total? What was the revenue?

Hint: Use a column chart. MEASURES – Revenue, DIMENSIONS – Product. Sort the Revenue, Highest to Lowest.

Question 11. Display the trend of revenue over rounds for each team.

Hint: Use a *line* chart. MEASURES – *Revenue*, DIMENSIONS– *Round and Day (level 2)*, TRELLIS or COLOR – *Team*.

Question 12. What is the market share of each team by product?

Hint: Use a Stacked Column chart 100%. MEASURES – *Revenue*, DIMENSIONS – *Product*, COLOR: *Team*

Question 13. Are there any products that do not sell in specific distribution channels? Explain.

Hint: Use a *Heat* map. DIMENSIONS – *Distribution Channel*, DIMENSIONS– *Product*, COLOR – *Quantity*.

Question 14. What was the highest priced muesli on day 18 of round 1? What team sold it? What was the price?

Hint: MEASURES – *Price*, DIMENSIONS – *Team, Product* and *SalesOrder*. Apply a filter.

Question 15. Which team sold the most quantity of muesli? For that team, what was the most sold product?

Hint: Use a Stacked bar/column chart. DIMENSIONS – *Team*, MEASURES – *Quantity*. COLOR - *Product*

Question 16. Show the days on which individual teams did not have any revenue.

Hint: Use a *Heat Map*. DIMENSIONS – *Round and Day*, DIMENSIONS – *Team*, COLOR - *Revenue*. Filter the results.

Question 17. What product on what day and round brought the highest revenue (for which team)?

No Hint!

9. Save.

All of the charts that you just created can now be used for creating presentations, dashboards, reports, and more. You can also share your story and your charts with others.

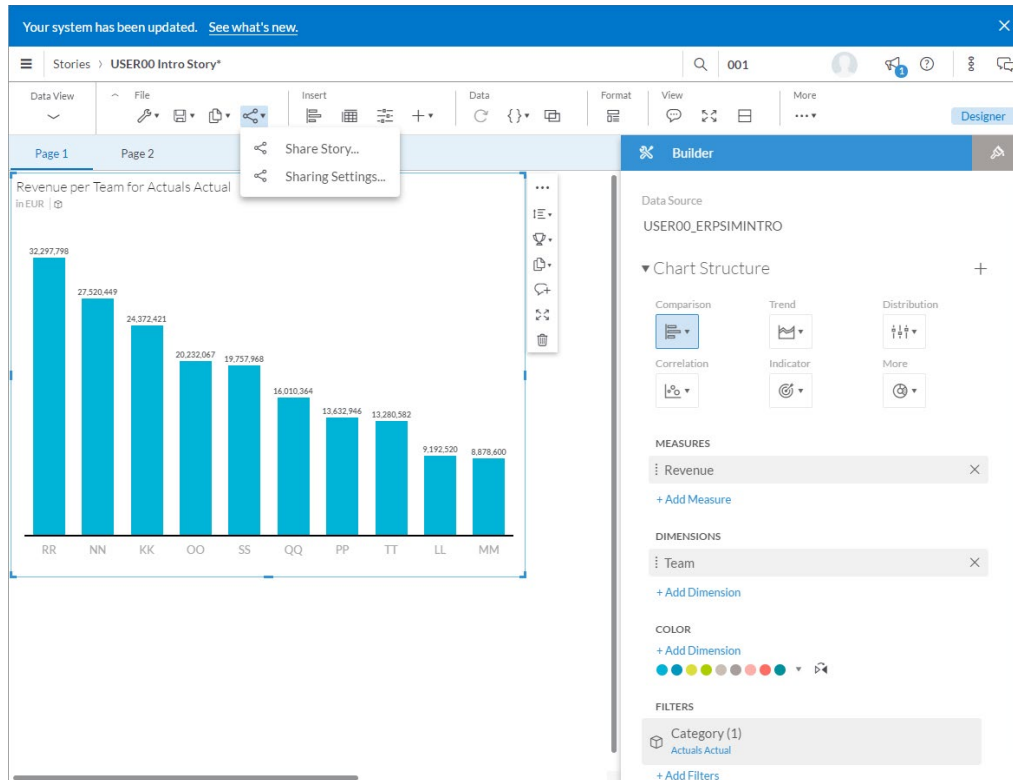


Figure 17: Sharing Your Story

10. Share → Share Story

- Share your story with your instructor using the dropdown list. Provide Full Control.

Share Story

Default Link
<https://higher-education.us10.sapanalytics.cloud/sap/fpa/ui/tenants/81639/bo/story/5E8100815D402A293781444E6661FEE8> [Customize Link](#)

Add Users or Teams

Access
 Full Control

☐ Apply global bookmark default
☐ Email new recipients

[Share](#) [Cancel](#)

> Shared with 1 User

[Close](#)

Figure 18: Share Full Control

Submission Requirements

- Provide answers to the questions on the answer submission document and include a screen shot of the visualization you used to answer the question. Screen shots can be created using the copy function on the visualization or by using a snipping tool.
 - Remember to share your Story with your instructor in SAC and be sure your name is in the title on each visualization.
 - Submit your assignment submission document by the due date and time.
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